



北京大学
PEKING UNIVERSITY

第六章 固体理论

3. 固体成键

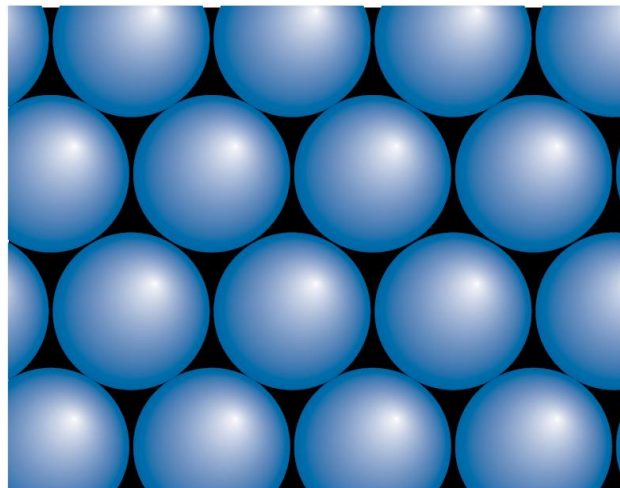
晶体分类：金属晶体、离子晶体、共价晶体、分子晶体

金属晶体：硬球模型

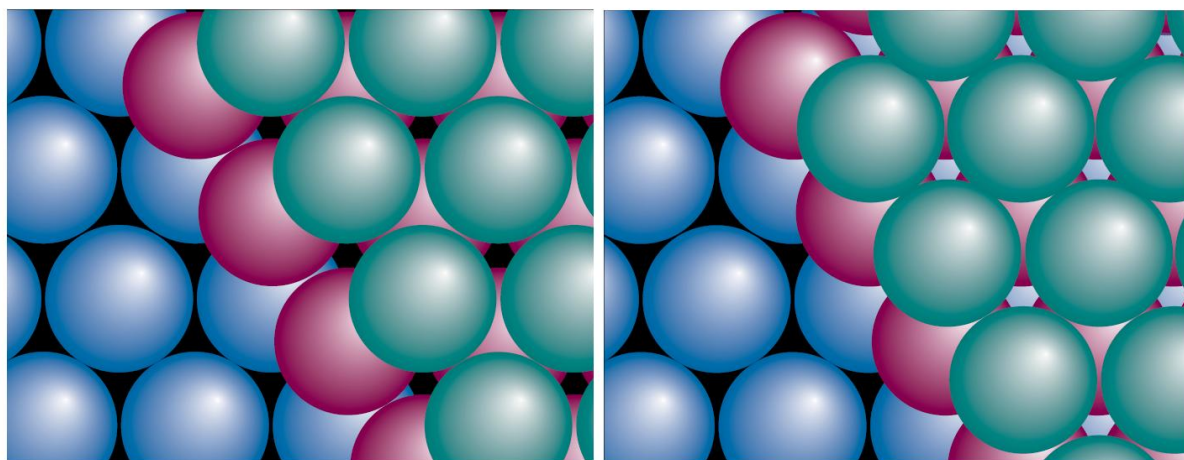
密堆积 (close packing)

三维空间密堆积

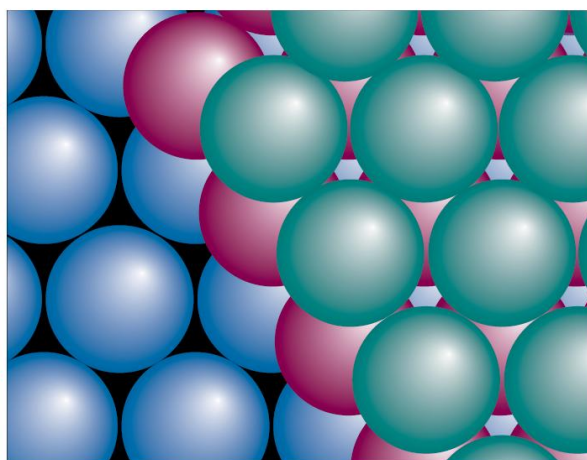
- 六方密堆积(hcp): ABABAB...
- 立方密堆积(ccp): ABCABC...



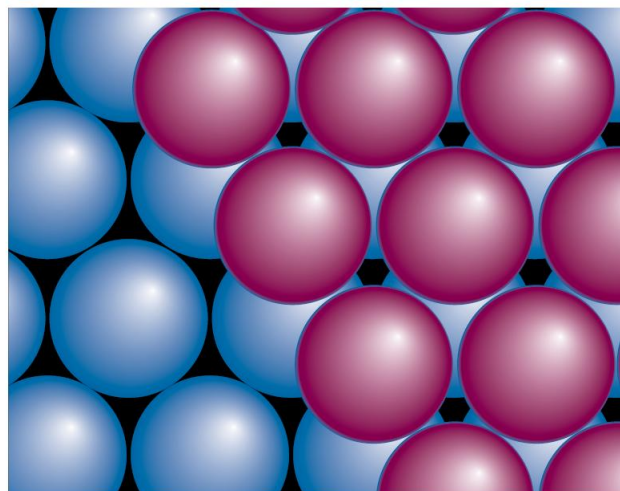
二维（单层）
密堆积



(a)

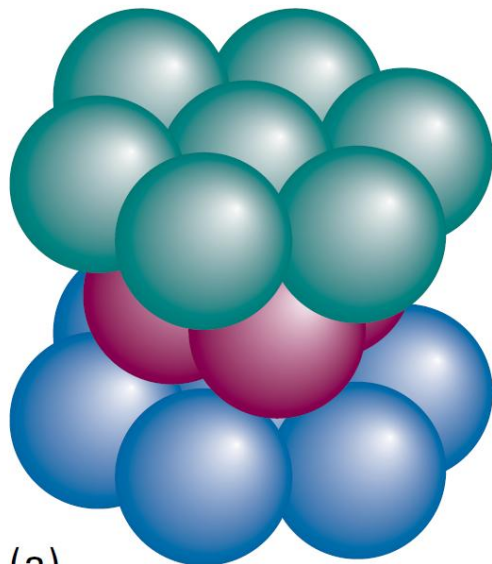


(b)



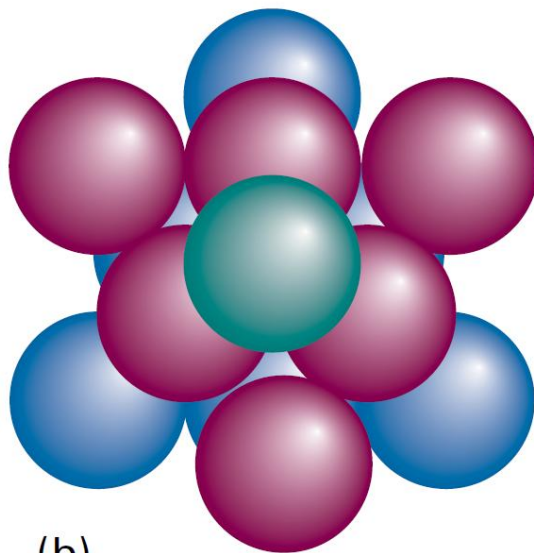
三维（双层）密
堆积，记为AB

- 六方密堆积(hcp):
ABABAB...



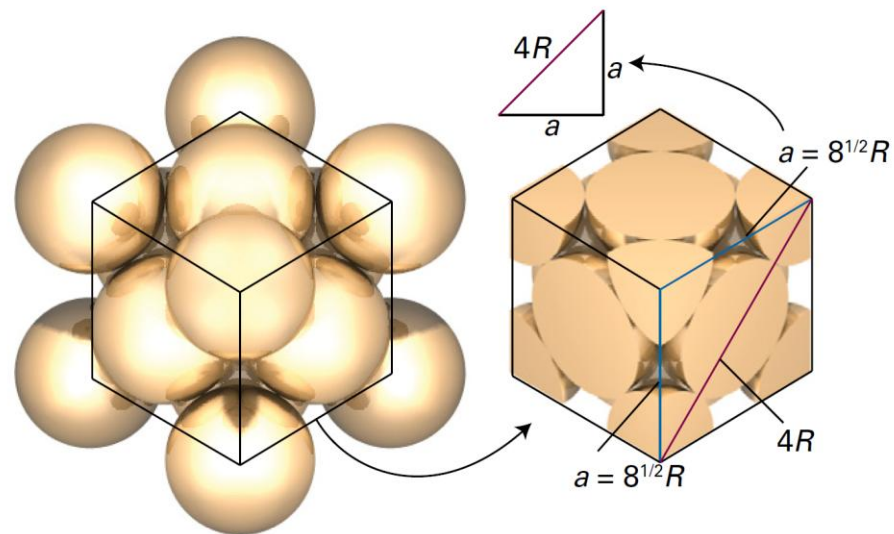
(a)

- 立方密堆积(ccp):
ABCABC...



(b)

空间占有率：都是0.74

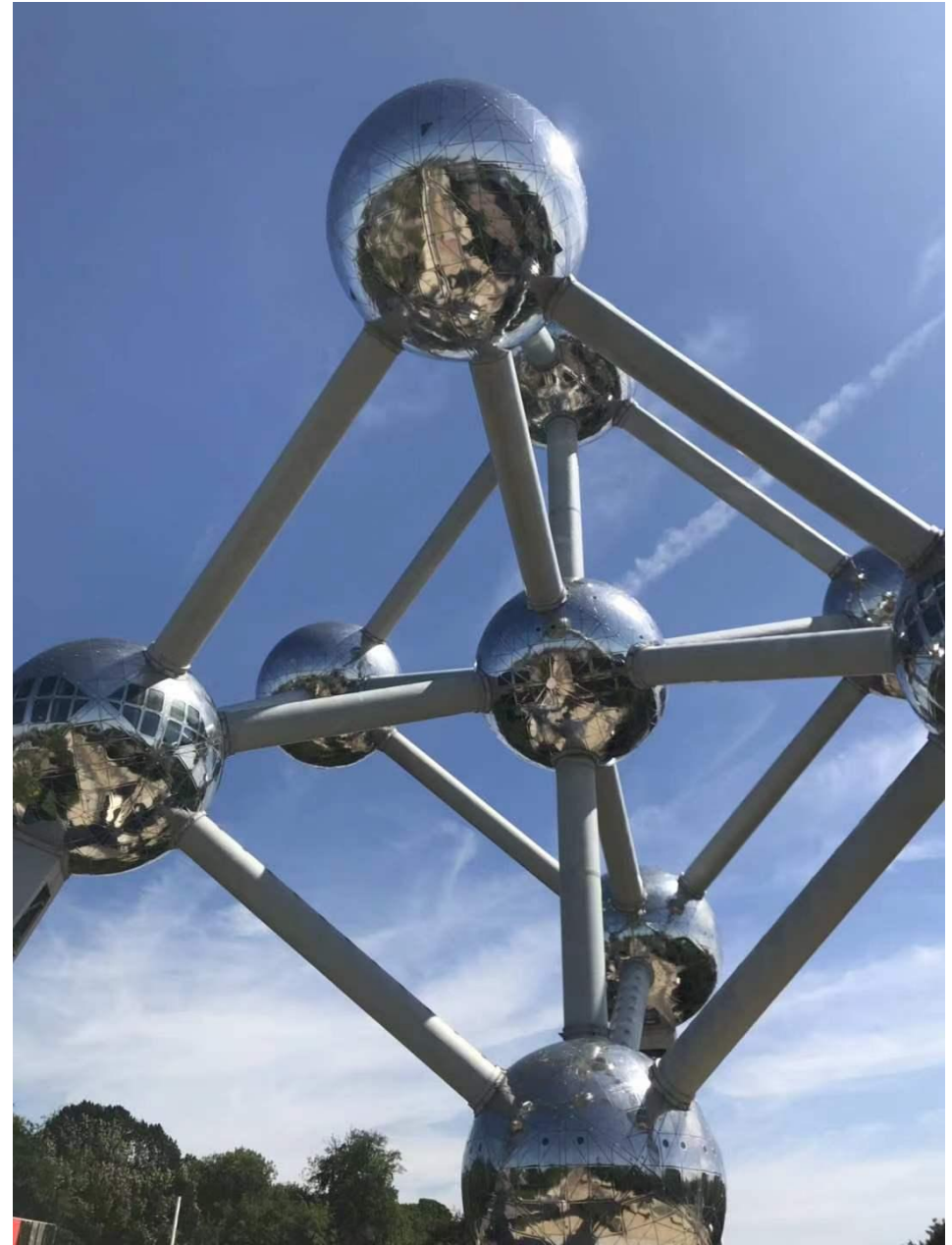


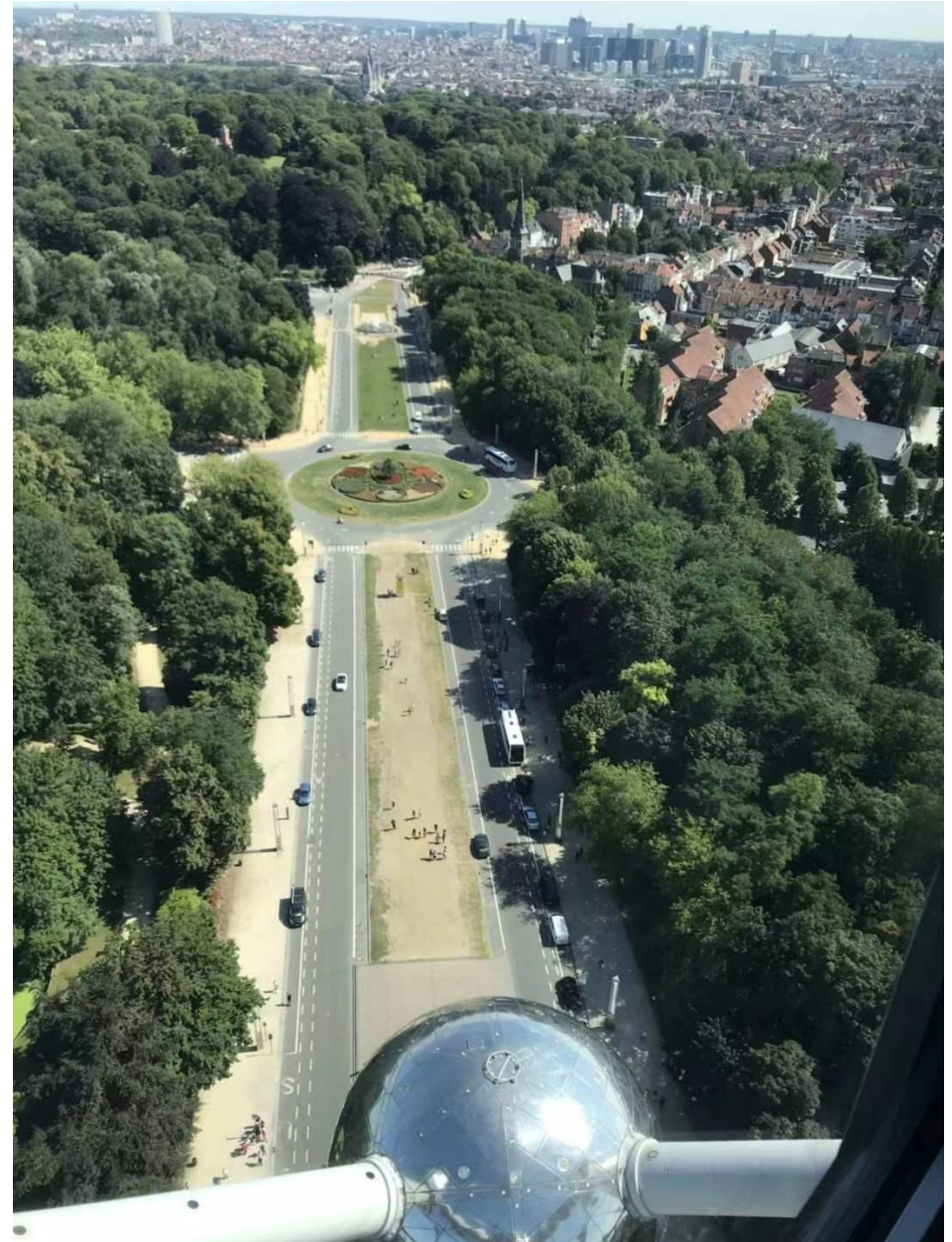
配位数都是12

ccp又叫fcc（面心立方）

其他可能性：ABC序列
随机，如ABACBAC...

Structure	Element
hcp [‡]	Be, Cd, Co, He, Mg, Sc, Ti, Zn
fcc [‡] (ccp, cubic F)	Ag, Al, Ar, Au, Ca, Cu, Kr, Ne, Ni, Pd, Pb, Pt, Rh, Rn, Sr, Xe
bcc (cubic I)	Ba, Cs, Cr, Fe, K, Li, Mn, Mo, Rb, Na, Ta, W, V
cubic P	Po





金属的电子结构（能带）

近自由电子近似

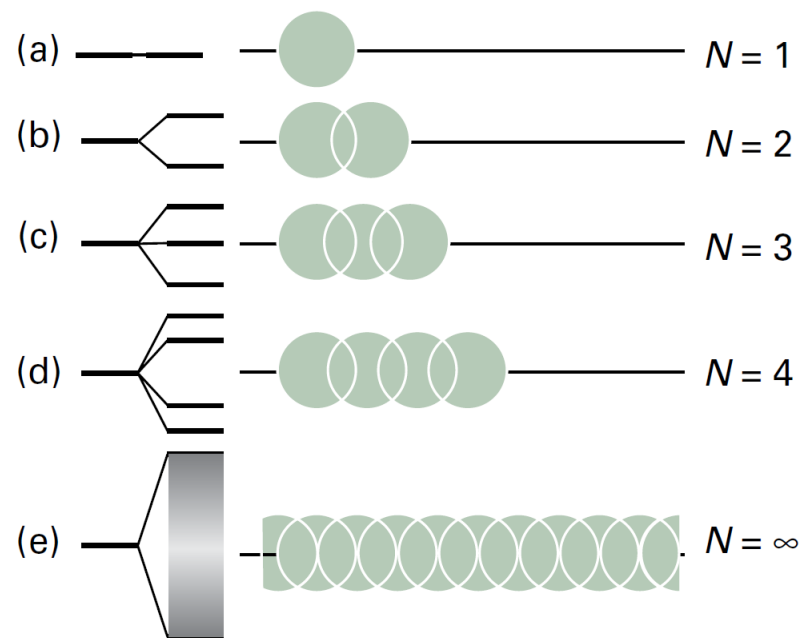
紧束缚（tight binding）近似

考虑一维周期性结构，每个原子贡献1s轨道组成分子轨道

休克尔模型计算能级

$$\begin{vmatrix} \alpha - E & \beta & 0 & \dots & 0 \\ \beta & \alpha - E & \beta & \dots & 0 \\ 0 & \beta & \alpha - E & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & \alpha - E \end{vmatrix} = 0$$

只考虑最近邻积分贡献



$$E_k = \alpha + 2\beta \cos \frac{k\pi}{N+1} \quad k = 1, 2, \dots, N$$

$E_N - E_1 \rightarrow -4\beta$ 随N增大，能级越来越稠密

$N \rightarrow \infty$, 形成能带

能带底：全成键

能带顶：全反键

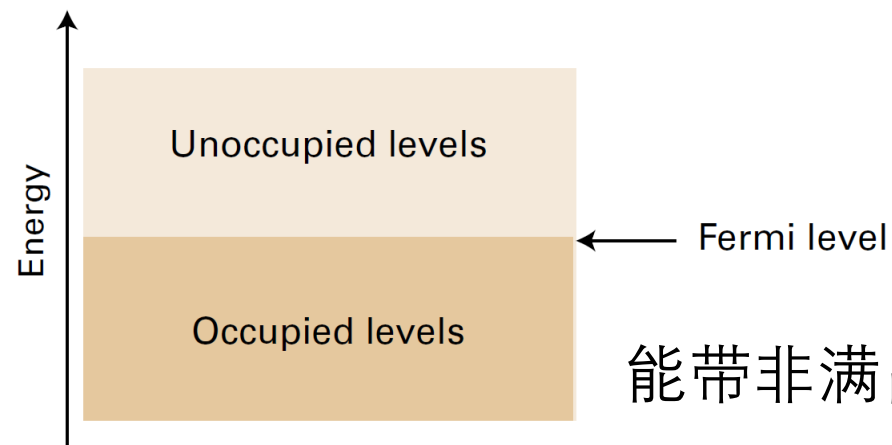
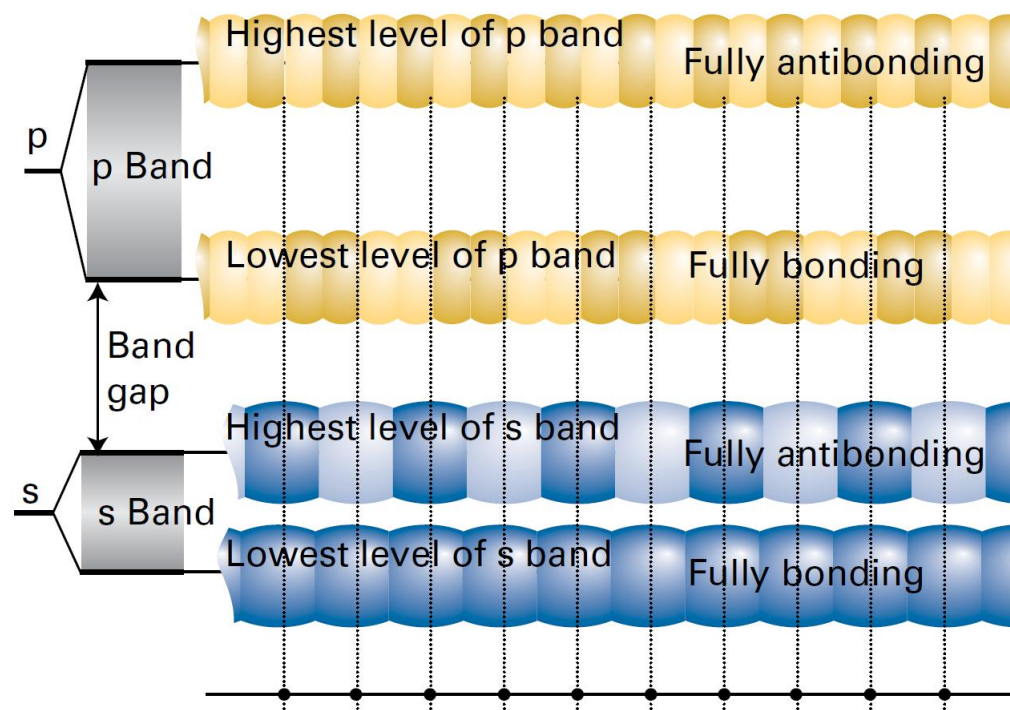
s轨道形成s带

p轨道形成p带

s带与p带间形成带隙
(band gap)

费米能级：HOMO ($T = 0$)

只有费米能级附近的电子
能被热激发 → 金属热容



能带非满占 → 导电性

离子晶体

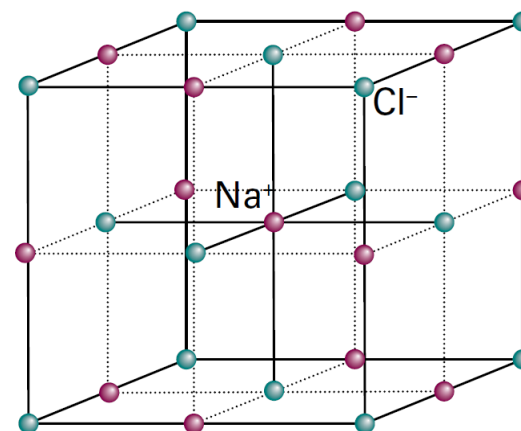
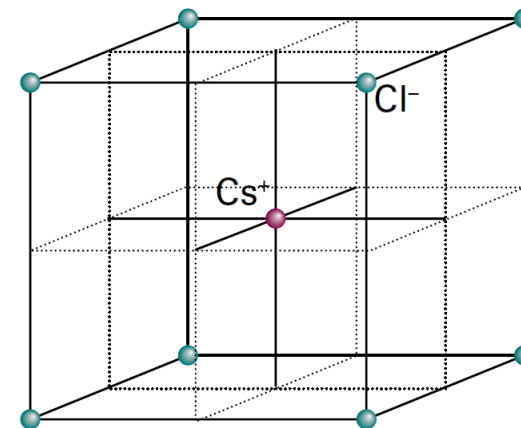
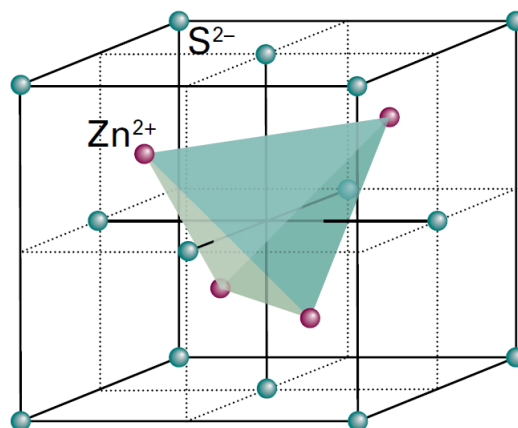
考虑阴阳离子半径区别

无法实现共同的密堆积，配位数<12

Radius ratio	Structural type
$\gamma < 2^{1/2} - 1 = 0.414$	sphalerite (Fig. 15C.11) 闪锌矿
$0.414 < \gamma < 3^{1/2} - 1 = 0.732$	rock salt (Fig. 15C.10)
$\gamma > 0.732$	caesium chloride (Fig. 15C.9)

$$\gamma = \frac{r_{\text{smaller}}}{r_{\text{larger}}}$$

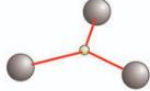
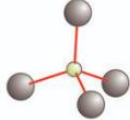
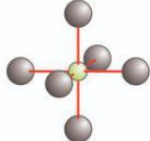
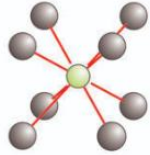
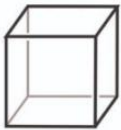
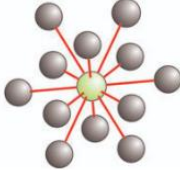
注意：阴离子构成多面体，阳离子在中心



配位多面体

空隙 (interstices)

R _c /R _a and Coordination of Cations		
R _c /R _a	expected coordination	coordination number
<0.15	2-fold coordination	2
0.15 0.15– 0.22	perfect triangular coordination triangular coordination	3
0.22 0.22– 0.41	perfect tetrahedral coordination tetrahedral coordination	4
0.41 0.41– 0.73	perfect octahedral coordination octahedral coordination	6
0.73 0.73– 1.0	perfect cubic coordination cubic coordination	8

coordination	bonds	polyhedron
linear (2-fold)		
triangular (3-fold)		
tetrahedral (4-fold)		
octahedral (6-fold)		
cubic (8-fold)		
dodecahedral (12-fold)		

配位多面体的连接方式

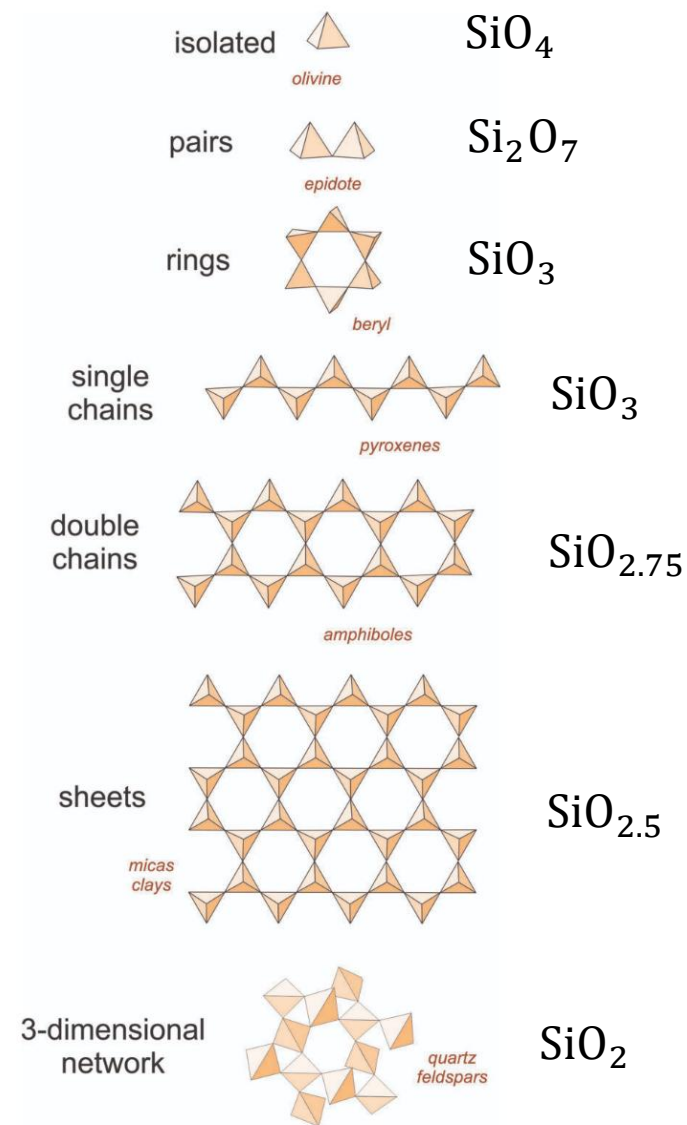
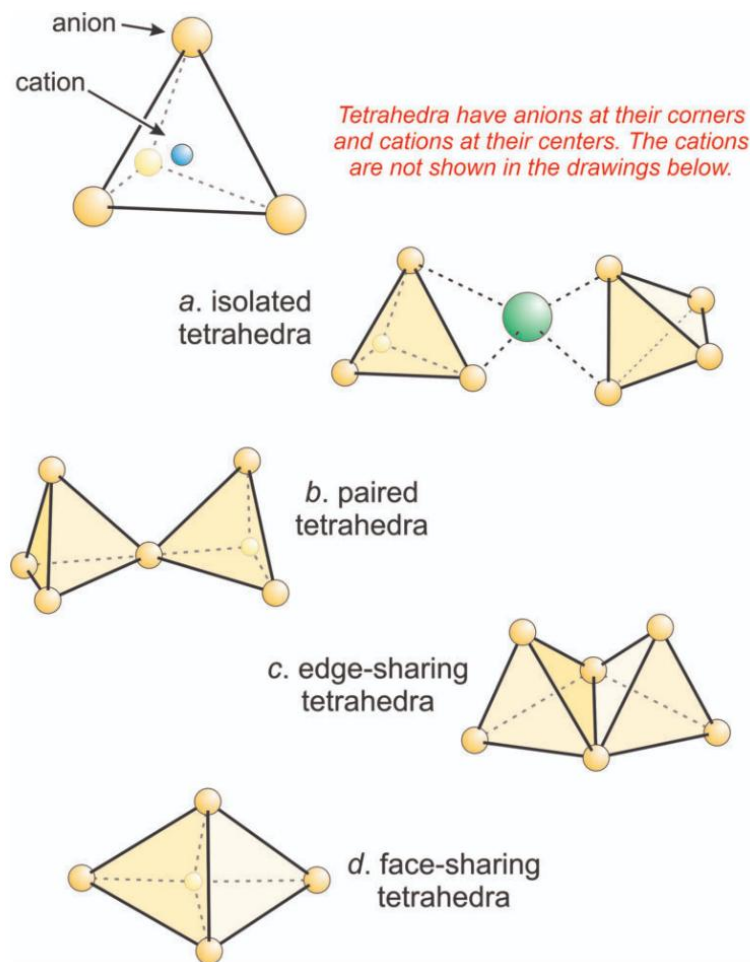
Si-O四面体

共顶点连接较多

共棱连接较少

共面连接更少

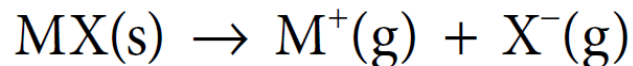
思考为什么?



Si-O四面体的空间排布方式

晶格能

固体由密堆积到气态的势能变化

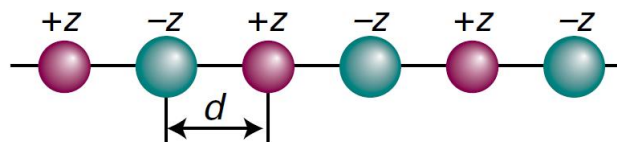


先考虑一维晶格，经典静电势能

$$\begin{aligned} E_p &= \frac{1}{4\pi\epsilon_0} \times \left(-\frac{z^2 e^2}{d} + \frac{z^2 e^2}{2d} - \frac{z^2 e^2}{3d} + \frac{z^2 e^2}{4d} - \dots \right) \\ &= \frac{z^2 e^2}{4\pi\epsilon_0 d} \times \overbrace{\left(-1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots \right)}^{-\ln 2} \\ &= -\frac{z^2 e^2}{4\pi\epsilon_0 d} \times \ln 2 \end{aligned}$$

$$E_p = -2\ln 2 \times \frac{z^2 N_A e^2}{4\pi\epsilon_0 d}$$

考虑两侧所有离子， $\times 2N_A$



推广到三维任意电荷

$$E_p = -A \times \frac{|z_A z_B| N_A e^2}{4\pi\epsilon_0 d}$$

A叫做Madelung常数

Structural type		A
Caesium chloride		1.763
Fluorite	萤石	2.519
Rock salt		1.748
Rutile	金红石	2.408
Sphalerite (zinc blende)		1.638
Wurtzite	纤锌矿	1.641

实际势能曲线

需考虑电子态能量贡献
(原子轨道交叠)

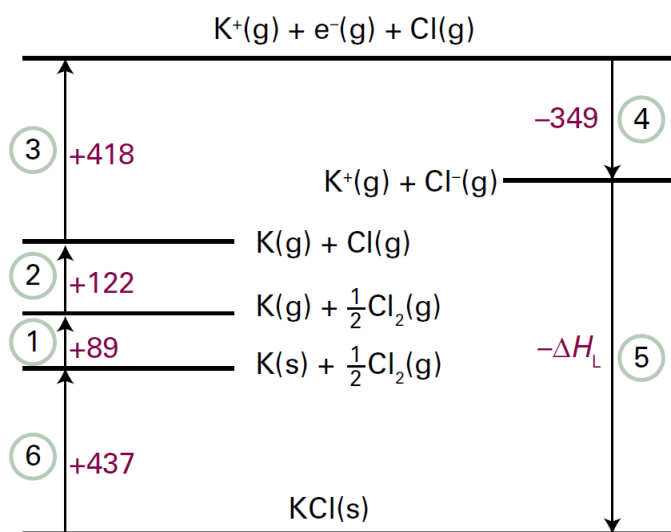
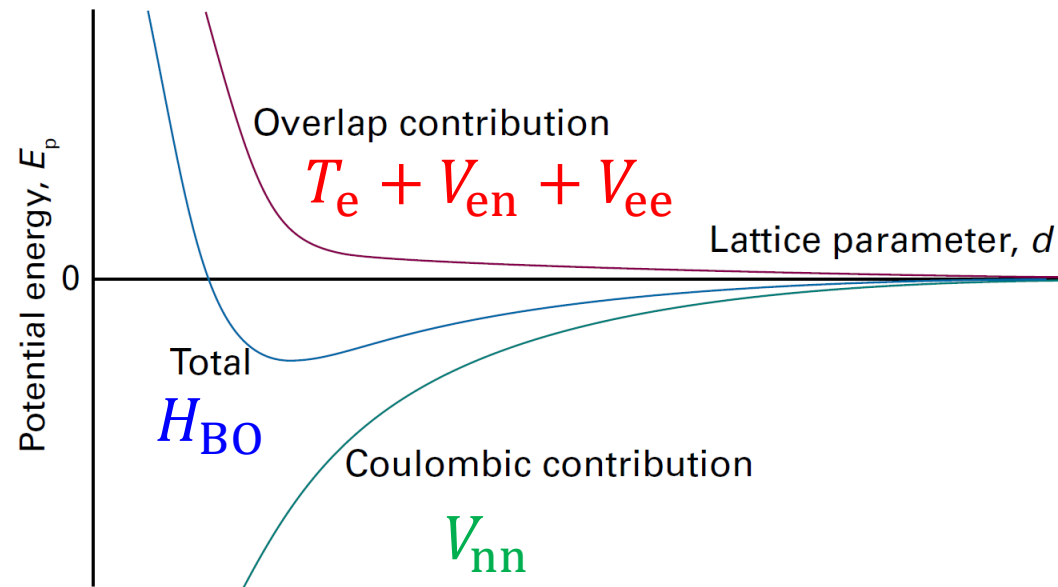
电子态能量贡献为正

经典势高估晶格能

$$E_{p,\min} = -\frac{N_A |z_A z_B| e^2}{4\pi\epsilon_0 d} \left(1 - \frac{d^*}{d}\right) A$$

高价离子，晶格能大

键长越小，晶格能越大



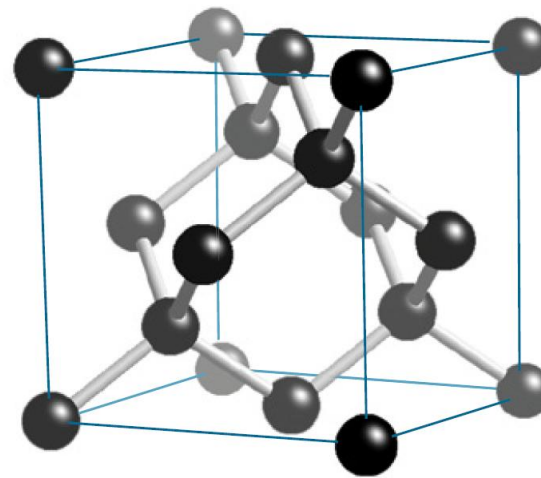
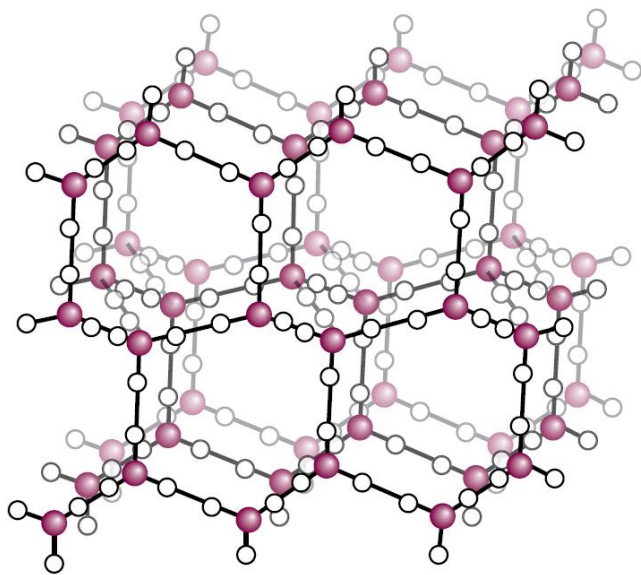
实验测定晶格能：
Born-Haber循环

共价晶体

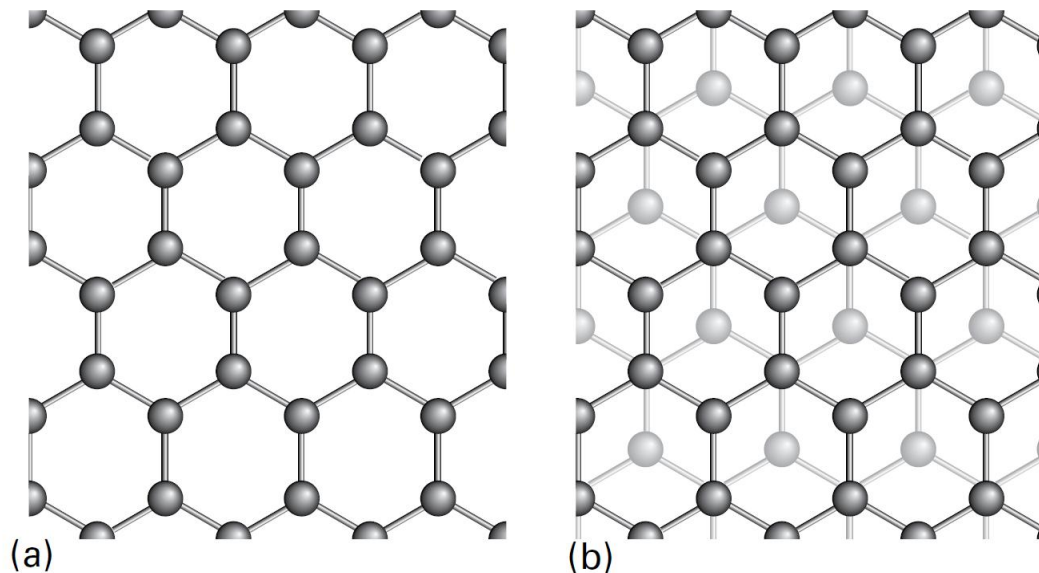
共价键方向性要求 > 密堆积要求

晶体结构复杂

分子晶体：通过范德华力凝聚



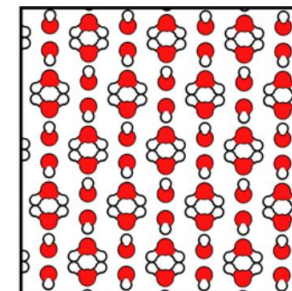
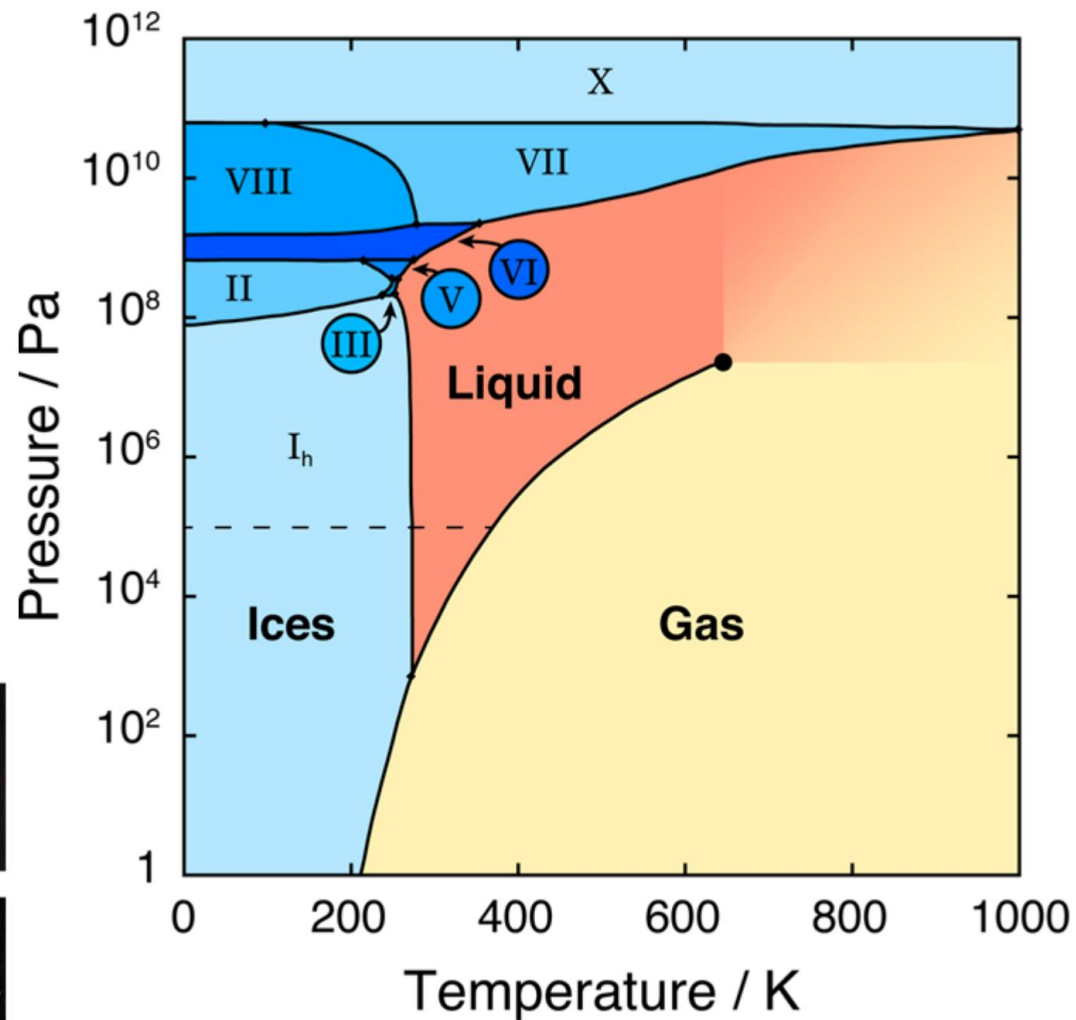
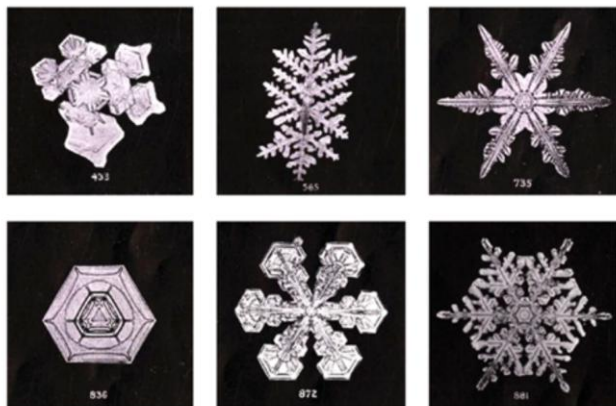
金刚石网络，硬度极强



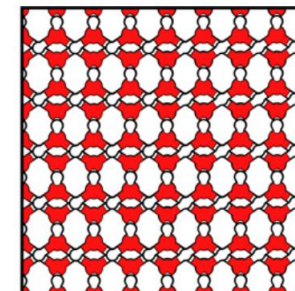
石墨层状结构，有杂质时用作润滑剂

水的相图

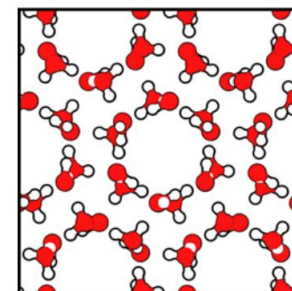
已知冰的17
种晶型 (图
中展示部分)



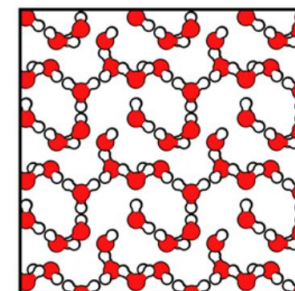
ice VIII 1.46 g/cm³



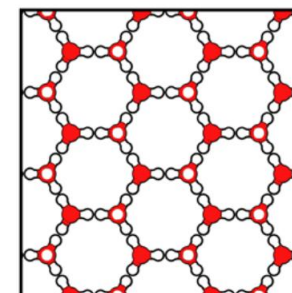
ice X 2.51 g/cm³



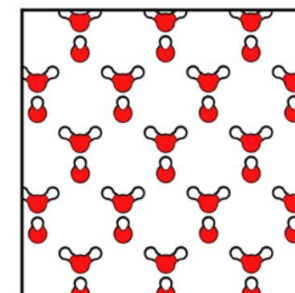
ice II 1.17 g/cm³



ice VI 1.31 g/cm³



ice I_h 0.92 g/cm³



ice I_c 0.93 g/cm³

E. Brini et al. Chem. Rev. 117, 12385, 2017.